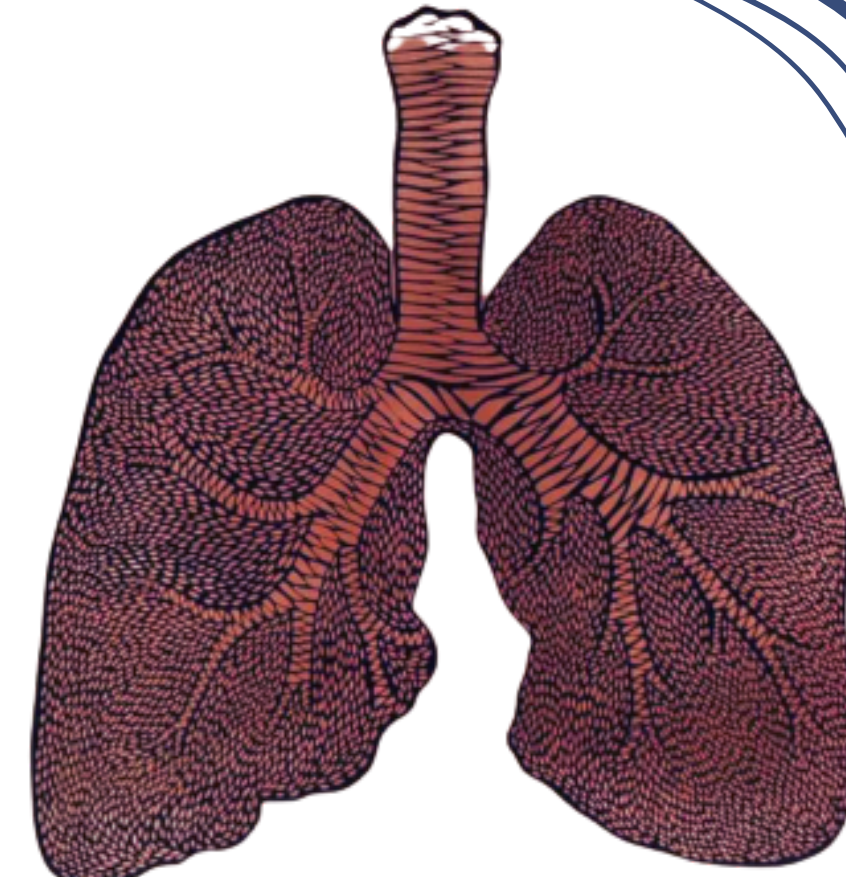


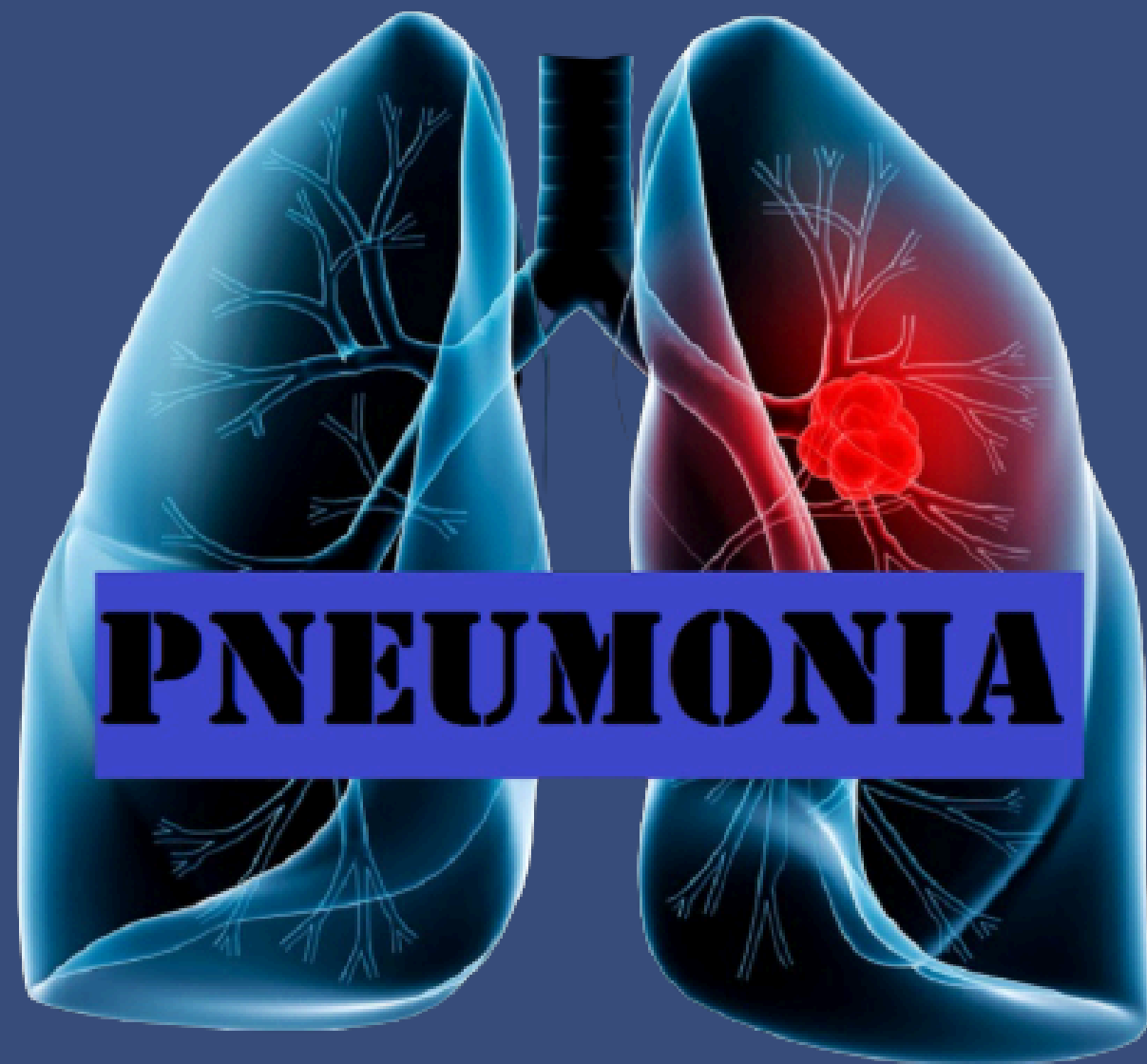
PNEUMONIA DETECTION USING OPTIMIZED DEEP LEARNING TECHNIQUE



Presented By

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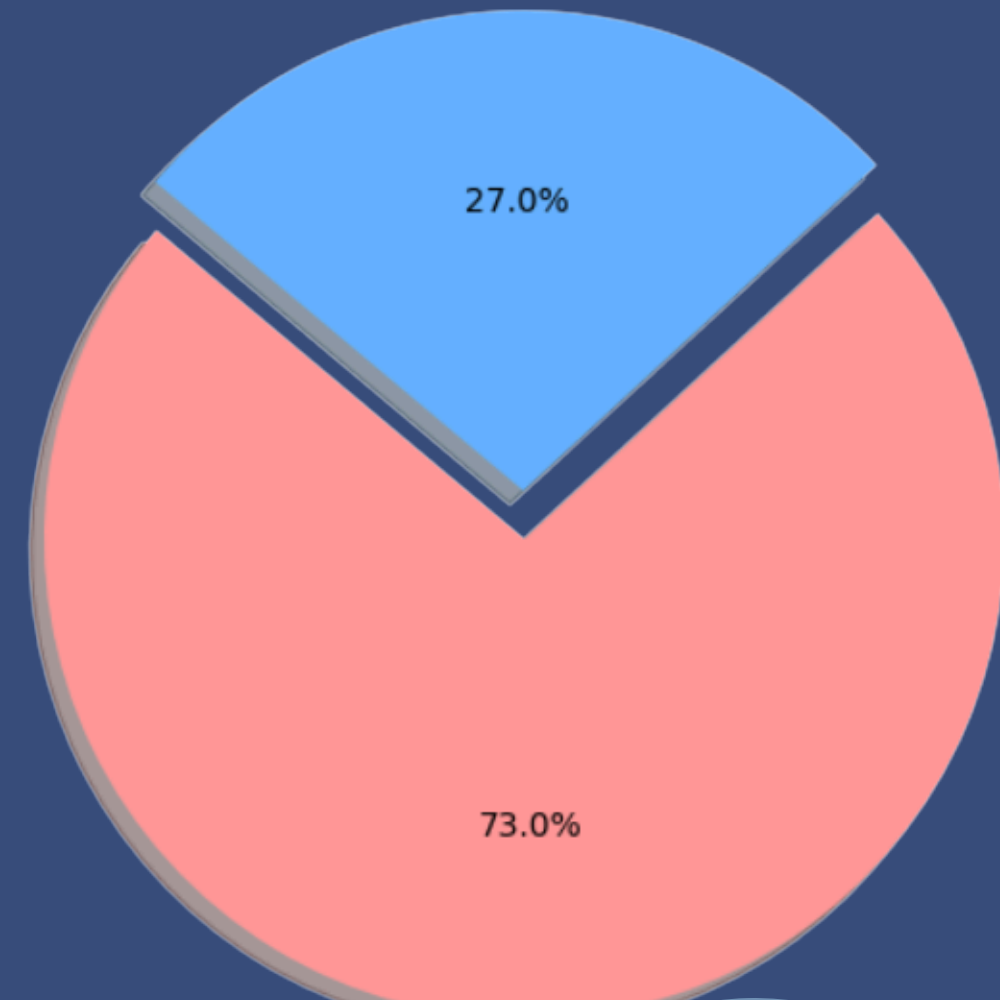
INTRODUCTION



- Problem Statement: Pneumonia is a serious respiratory infection that requires early detection for effective treatment.
- Objective: Develop a deep learning model to classify chest X-ray images into "Normal" or "Pneumonia" cases.
- Importance: Automating detection can assist doctors in diagnosing pneumonia faster and more accurately.

DATASET OVERVIEW

- Source: Chest X-ray images from the dataset directory(KAGGLE).
- Class Distribution:
 - Normal: 1,584 images
 - Pneumonia: 4,273 images



DATA PREPARATION



1. Directory Structure:

- Dataset divided into "NORMAL" and "PNEUMONIA" folders.

2. Image Loading:

- Images loaded using TensorFlow's ImageDataGenerator for preprocessing.

3. data augmentation:

- added augmented normal images to remove the imbalance of classes.

TRAINING PROCESS

- Data Augmentation: Applied to increase dataset diversity (rotation, zoom, etc.).
- Evaluation Metrics:
 - Confusion Matrix: To evaluate true positives/negatives.
 - Classification Report: Precision, recall, and F1-score.

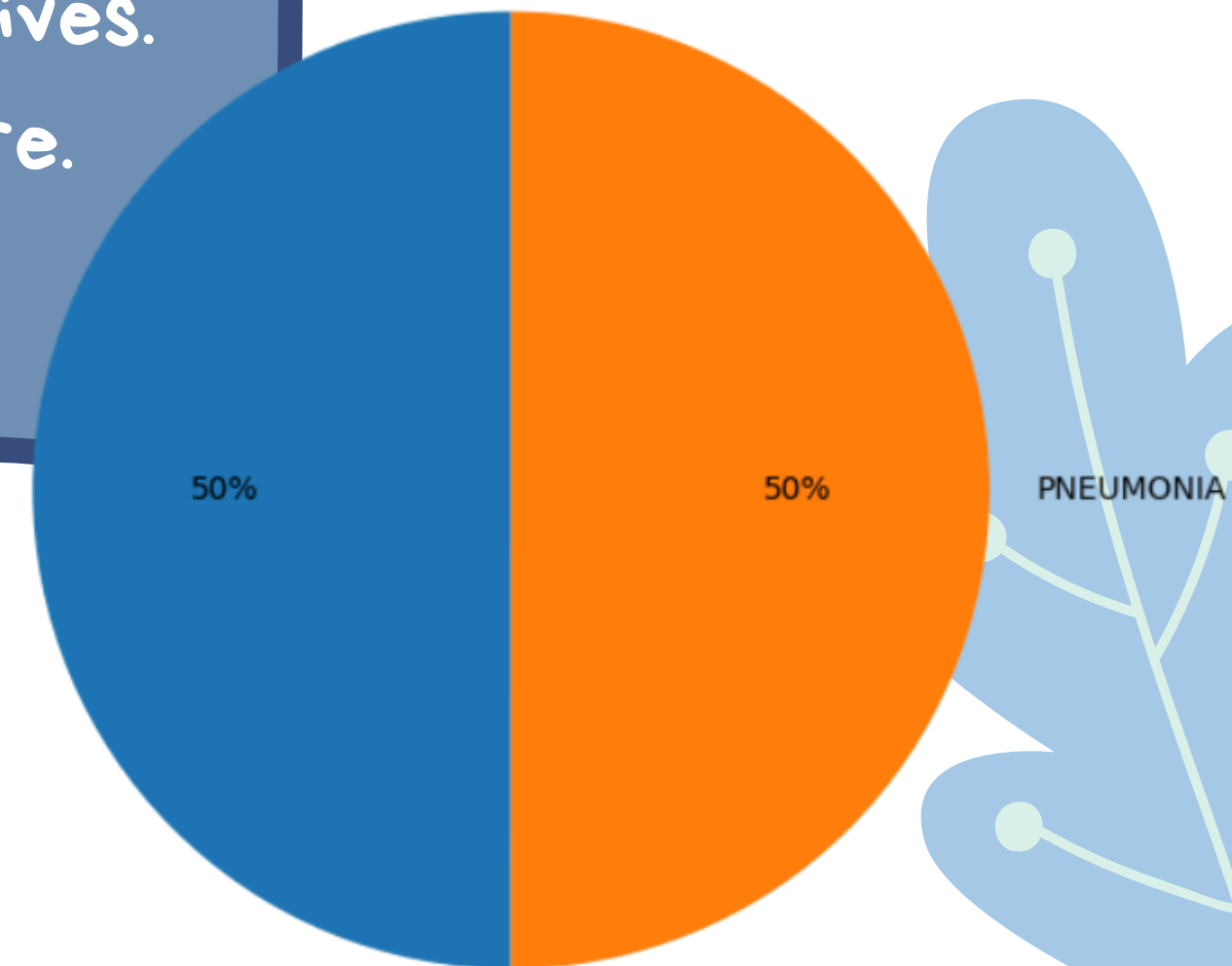
Training Dataset

NORMAL

50%

50%

PNEUMONIA



MODEL ARCHITECTURE

- Model Type: Sequential CNN (Convolutional Neural Network).
- Layers:
 - Conv2D: Feature extraction.
 - maxpool: focus on the important features
 - Flatten: Convert 2D features to 1D.
 - Dense: Fully connected layers for classification.
 - Transfer Learning: Utilized VGG16 for improved accuracy.

METHODOLOGY

1. Baseline CNN:

- Started with a standard Sequential CNN (Conv2D, Flatten, Dense layers).
- Trained on chest X-ray images but faced suboptimal accuracy due to model simplicity.

2. Iterative Copy & Optimization:

- Copied and modified the baseline CNN multiple times, experimenting with:
 - Different layer architectures (added custom layers like BatchNorm, Dropout).
 - Hyperparameter tuning (learning rate, filters, kernel sizes).
 - Transfer learning (VGG16 fine-tuning).
- Goal: Maximize accuracy by "evolving" the best-performing network.

3. Introducing Metaheuristics:

- Applied Genetic Algorithm (GA) to train the last added layers:
 - by mutation and selection.
 - freezing the imported pretrained model weights
- Compared against other algorithms:
 - Particle Swarm Optimization (PSO) for weight optimization.
 - Ant Colony Optimization (ACO) for feature selection.
 - Grey Wolf Optimizer (GWO)
 - Bayesian Optimization (BO).

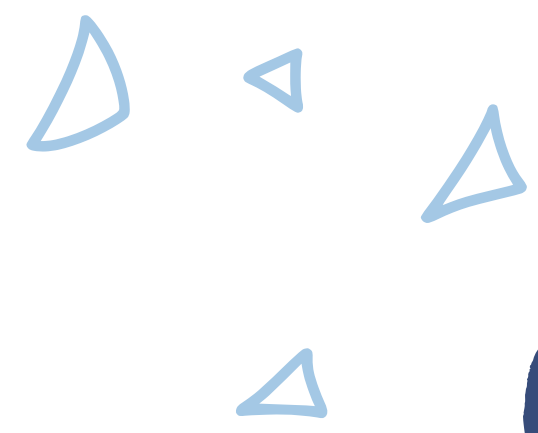
ADVANCED MODEL ARCHITECTURES (LEVERAGING YOLOV8 & VGG16 FOR ENHANCED PERFORMANCE)

Why These Models?

- VGG16: Proven for image classification (pre-trained on ImageNet).
- YOLOv8: State-of-the-art for object detection (localizes pneumonia regions).

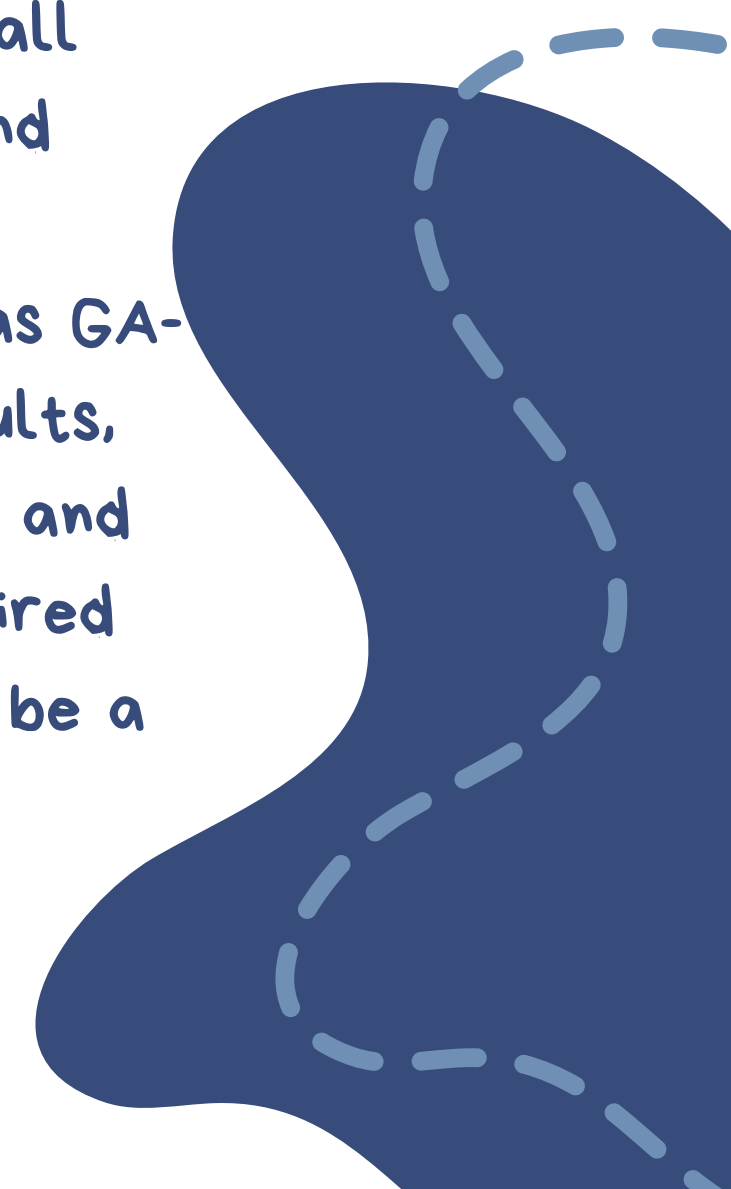
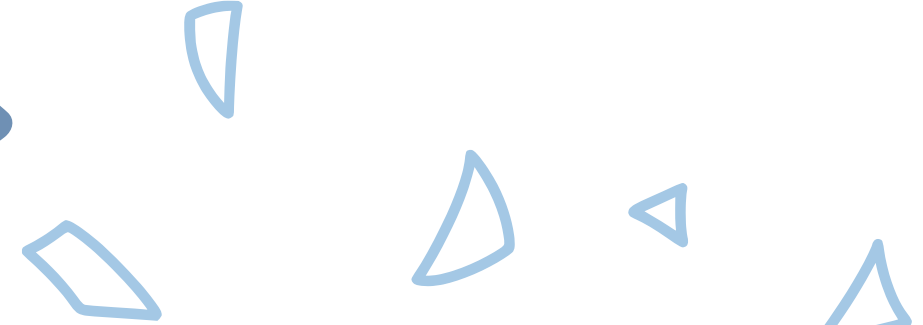
RESULT

Model/Optimization	Accuracy	Precision	Recall	F1-Score	Key Advantages
Baseline CNN before training	49.96%	0.49	1.0000	...	
Baseline CNN after training	93.07%	0.93	0.92	...	Simple implementation
GA-(DEAP)	95.87%	0.99	0.93	0.96	Best lesion detection + classification
GA-Optimized PyGad	96.57%	0.96	0.97	0.97	Balanced architecture optimization
Grey Wolf Optimizer (GWO)	96.41%	0.98	0.94	0.96	Effective feature selection
Ant Colony (ACO)	96.65%	0.97	0.96	0.97	Good for path optimization
Particle Swarm (PSO)	97.43%	0.98	0.97	0.97	Weight tuning
Bayesian Optimization	96.41%	0.98	0.95	0.96	Hyperparameter search efficiency



CONCLUSION

The comparative analysis highlights the significant impact of optimization algorithms on improving the performance of deep learning models for medical image classification. While the baseline CNN model achieved poor results with only 36.51% accuracy, the use of optimization techniques led to substantial performance boosts. Among all methods, the PSO-Optimized model achieved the highest accuracy (97.43%) and excellent precision, recall, and F1-score (0.98, 0.97, and 0.97 respectively), demonstrating its strength in fine-tuning model weights. Other approaches such as GA-Optimized Hybrid, ACO, and Bayesian Optimization also showed competitive results, each offering unique advantages like architecture balancing, path optimization, and hyperparameter efficiency. This clearly indicates that integrating nature-inspired optimization methods significantly enhances classification performance and can be a crucial step toward more accurate and reliable medical diagnosis systems.





Thank You